



Investigations of hemato-biochemical and histopathological parameters, and growth performance of walking catfish (*Clarias batrachus*) exposed to PET and LDPE microplastics

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ABSTRACT

Fish inhabiting various trophic levels are affected differently as the presence of microplastic (MP) in the water column and their ingestion by fish varies. Walking catfish (*Clarias batrachus*) inhabits the bottom of the water bodies. To understand the effects of MP, we exposed *C. batrachus* to two types of MP – polyethylene terephthalate (PET) and low-density polyethylene (LDPE) for 60 days. After exposure, hematological indices, mainly red blood cells and hemoglobin levels decreased, and white blood cells increased significantly compared to the control group ($p < 0.05$). A significant increase in the levels of blood urea and glucose was observed, and serum glutamic pyruvate transaminase and serum glutamyl oxaloacetic transaminase activity remained elevated ($p < 0.05$). Histopathological examination of the liver, kidney, intestine, and gills showed morphological alterations. Moreover, MP exposure caused growth retardation ($p < 0.05$) in *C. batrachus*. Widespread pollution of water bodies by MP may impose serious ecological risks to bottom-feeding fish in Bangladesh.

1. Introduction

Plastics are polymerization-based monomers with distinct physico-chemical properties and a durable structure that can persist for hundreds of years and pose a severe threat to global contamination (Hamed et al., 2022; Jiang et al., 2018). It is projected that, within a few decades, the amount of plastic entering the marine environment will rise to the level that it will exceed the ocean's fish mass (Simon and Schulte, 2017). Worldwide plastics production rose to 390.7 million tonnes in 2021 after stagnation in 2020 owing to the Covid-19 outbreak (Plastics Europe, 2022). Plastics that enter seas and oceans are converted to macroplastics, mesoplastics, and microplastics following chemical and biological reactions, including solar degradation, biodegradation, etc. (Jambeck et al., 2015). Due to the small size (1 μm to 5 mm) and global occurrence in coastal-marine waters (Alimba and Faggio, 2019; Amelia et al., 2021; Fatema et al., 2023; González-Ortegón et al., 2022; Napper et al., 2021; Sutton et al., 2016), microplastic (MP) are readily available

to different aquatic biota in all water columns (Foekema et al., 2013; Mathalon and Hill, 2014).

The use of manmade synthetic polymeric materials plays a significant influence on our everyday life. Synthetic polymers, including polyethylene (PE), polyethylene terephthalate (PET), polyvinyl chloride (PVC), nylon, etc., are used in a wide range of sectors, including packaging, textiles, paper, automobile, etc. (da Costa et al., 2016; de Sá et al., 2018; Jiang et al., 2019). Particularly, PE [(C₂H₄)_n], also categorized as LDPE – low-density PE, HDPE – high-density PE and LLDPE – linear low-density PE, is used in a variety of products, including plastic bags, sheets, threads, packaging, toys, and other things (Lusher et al., 2017; Fatema et al., 2023). PET [(C₁₀H₈O₄)_n] serves as the primary ingredient in textiles, water bottles, the majority of colorful fibers, several transparent components, etc. (Fatema et al., 2022; Iñiguez et al., 2017; Wang et al., 2017). Given the fact that plastics can persist in the ecosystem for an extended period, MP and their environmental effects are unavoidable (Cole et al., 2011). MP have been found to be ingested by various

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organisms throughout the world (Clere et al., 2022; Digka et al., 2018; Franzellitti et al., 2019; Hara and Nash, 2020; Jabeen et al., 2017; Li et al., 2016; etc.). Adverse effects of MP on the physiology and health of different aquatic organisms have been reported worldwide (Abarghouei et al., 2021; Critchell and Hoogenboom, 2018; Ding et al., 2018; Hamed et al., 2022; Impellitteri et al., 2023; Jabeen et al., 2018; Raza et al., 2023), yet there are still rising concerns with regards to health and survival of fish in the aquatic ecosystem and human health due to possible trophic transfer of MP via food chain (Borgstrom, 1961; Chen et al., 2022; Jurek et al., 2022).

The walking catfish *Clarias batrachus* (Linnaeus, 1758), a species belonging to the family Clariidae and the order Siluriformes (Ng and Kottelat, 2008), is one of the most popular freshwater fish in Bangladesh. It can be found in freshwater and brackish waters of India and adjacent South Asian countries (Talwar and Jhingran, 1991). *C. batrachus* is a suitable model for studying MP toxicity as these fish are readily available, have wider tolerance to various environments, and have different food habits. This fish is frequently used to examine various harmful toxicological effects (Gupta and Verma, 2022; Khan and Badroo, 2021; Koner et al., 2021; Kumar and Banerjee, 2016; Murganankumar et al., 2016; Saha et al., 2023; Soni and Verma, 2018).

Hematological and histological parameters can be used to assess environmental contamination (Abarghouei et al., 2021; Fazio, 2019; Jabeen et al., 2018; Kim et al., 2021; Sayed and Authman, 2018; Suchana et al., 2021). Additionally, biochemical indices are considered to be helpful bioindicators for determining the effects of various contaminants (Hamed et al., 2019, 2022; Impellitteri et al., 2022; Kavitha et al., 2012; Kumar and Banerjee, 2016; Nematdoost Haghi and Banaee, 2017; Sayed and Soliman, 2018). Moreover, the histological analysis offers a very affordable and valuable ecological risk assessment method while non-specific (van der Oost et al., 2003). To our knowledge, the impact of PET fragments and LDPE pellets on the freshwater walking catfish has not yet been revealed. Thus, the present study aims to determine the changes in hemato-biochemical parameters and histopathological alterations in internal organs of *C. batrachus* as well as their growth performance caused by MP with different types and chemical composition (i.e., PET fragments and LDPE pellets). The findings of this study might contribute to our growing understanding of MP's effects on aquatic ecosystems.

2. Materials and methods

2.1. Experimental fish and design

Healthy, active, and disease-free walking catfish (*Clarias batrachus*) of an average total length of 23.03 ± 2.63 cm and average body weight of 103.40 ± 3.26 g were collected from a nearby hatchery named Sharnalata Agro Fisheries Limited, Fulbaria, Mymensingh. They were housed in a cemented rectangular cistern (length: 250 cm, width: 195 cm, and height: 70 cm) with a continuous aeration system for 30 days to acclimate at the Faculty of Fisheries, Bangladesh Agricultural University, Bangladesh. The fish were fed commercial feed (4–3% of body weight) once daily at 9.00 am, and the water temperature was kept at 27 °C by a thermostatically controlled system. After acclimatization, apparently, healthy fish were selected for MP exposure in a wet laboratory. A total of 240 fish [20 fish per tank $\times$ 3 replicates = 60 fish per group] were stocked in twelve cleaned tanks (diameter: 170 cm, and height: 75 cm) and filled each with 350 L of clean tap water. The use of walking catfish (*Clarias batrachus*) in this study was approved by the Ethical Standard of Research Committee of the Bangladesh Agricultural University Research System.

The four experimental treatments were T0 control (commercial feed only), T1 fed with commercial feed + PET fragments (fed separately), T2 fed with PET fragment-mended commercial feed, and T3 fed with LDPE pellet-mended commercial feed. In the supplementary file, different diets and details on the experimental design scheme have been

presented in Fig. S1 and Fig. S2, respectively. Adequate aeration was maintained at each tank throughout the testing period using a thermostatically controlled system. The fish were fed (3% of body weight) once daily at 9.00 am. The uneaten food was removed after 1 h. We changed the tanks' water regularly throughout the experiment. The water quality parameters (dissolved oxygen, pH, alkalinity, and free CO₂) were measured and maintained in a suitable range for fish (Fig. S3). pH was determined using a pH meter (Model: RI 02895, HANNA); dissolved oxygen was measured using a DO meter (Model: DO- 5509, Lutron, Taiwan); alkalinity was measured by titration method (0.02 N standard sulphuric acid as titrant and methyl orange as an indicator); free carbon dioxide was measured by titration method (0.0227 N sodium hydroxide as titrant and phenolphthalein as an indicator).

2.2. MP-mended feed preparation

Fragments and pellets (<5 mm) were used in this study. Fragments were irregular, green in color, and ranged from 1 to 2 mm, while pellets were round and white and ranged from 2.9 to 3.0 mm in size. The chemical compositions of plastic items were determined by Fourier Transform Infrared Spectroscopy (FTIR-4600, JASCO Inc., Japan) to be polyethylene terephthalate (PET) fragments and low-density polyethylene (LDPE) pellets (Fig. S4).

MP-mended feeds were prepared using the method outlined by Jabeen et al. (2018) with some modifications. Virgin LDPE pellets and PET plastic bottles were collected from the local market. The plastic bottles were chopped into small fragments with the help of scissors and a scalpel (Roch et al., 2020). Commercial fish feed (Magur grower floating feed) was softened by soaking it in clean water. Then pellets or fragments were added to the softened feed and dried in sunlight with a clean net covering. PET fragments mended feed was prepared by weight (each 1 g feed contained 0.05 g of fragments). Pellet-mended feed was prepared using particle number (each food particle contained one pellet). In the current study, we apply both voluntary and forced feeding techniques and provided 3–6 fragments/L in the case of T1 and T2, whereas 1–2 pellets/L in the case of T3 considering environmentally relevant particle concentrations (Hossain et al., 2022; Min et al., 2023; Napper et al., 2021; Qin et al., 2023; Weideman et al., 2020; Yuan et al., 2019). The control feed particles had no plastic items. The feed was stored in airtight glass jars in the laboratory at room temperature for daily feeding.

2.3. Hematobiochemical study

On days 30 and 60 of dietary MP exposure, 9 fish from each group were taken and anesthetized with 5 mg/L of clove oil. Blood samples were collected with a sterile, non-heparinized syringe through a caudal vein puncture and left for 30 min for coagulation. Serum samples were collected after centrifuging at 4000 rpm for 15 min at 4 °C (Eppendorf Centrifuge HERMLE Z 216 MK) and then kept in ice to determine biochemical indices. Afterward, glucose, urea, triglycerides, total protein, serum glutamyl oxaloacetic transaminase (SGOT) [aspartate transaminase (AST)] and serum glutamic pyruvate transaminase (SGPT) [alanine transaminase (ALT)] activities were analyzed using a semi-auto chemistry analyzer (Biogen–5500).

The blood sample was drawn from the caudal peduncle of each fish and stored in Eppendorf tubes with 20 mM EDTA anticoagulants until red blood cells (RBCs) and white blood cells (WBCs) were counted. RBCs and WBCs counts were performed with the Neubauer hemocytometer under a digital microscope (HumaScope Classic^{LED}). Furthermore, blood hemoglobin (Hb) levels were measured using hemoglobin strips in a digital Easy Touch® GCHb.

2.4. Histopathological investigation

Studies revealed MP have been identified in different organs of fish

(gills, intestine, liver, etc.) and cause physiological damage (Abarghouei et al., 2021; Hamed et al., 2021; Jabeen et al., 2018; Lei et al., 2018; Pedà et al., 2016; Roch et al., 2020). In the present study, we chose the gills, intestine, liver, and kidney for histopathological analysis.

The liver, intestine, kidney, and gills were collected from each sampled fish ($n = 9$) on days 30 and 60 after dietary MP exposure. The tissues were fixed in Bouin's fluid for 24 h. The tissue samples were stored in 70% alcohol and kept at 4 °C until histopathological analysis. The fixed tissues were dehydrated through a series of increasing alcohol concentrations, according to standard protocols. Tissues were blocked in wax and sectioned at 5 μm thickness. A total of 45 slides were made per sample.

Then, the slides were treated with xylene for 5 min to remove wax, followed by concentrations of alcohol. Sections were stained with Hematoxylin and Eosin. All the prepared slides were examined under a digital microscope (HumaScope Classic^{LED}) and photographed (AmScope, Model: MU 1003).

2.5. Analysis of growth performance

Body weights were estimated for all fish at the start and end of the experiment. Fish mean weight was calculated for each replication tank by dividing the overall weight by the total number of fish. The ensuing growth indices for 2 months (60 days) were computed. The following equations were used to determine morphometric parameters: Fish weight gain (%) = [(mean final weight of fish – mean initial weight of fish)/mean initial weight of fish] $\times$ 100. Specific growth rate (% day⁻¹) = [(Ln final body weight – Ln initial body weight)/duration of experiment (days)] $\times$ 100. Survival rate (%) = [Total fish survived at the time of sampling/Total fish stocked initially] $\times$ 100.

2.6. Statistical analysis

For hematological and serum biochemical parameters, descriptive analyses were performed to determine mean $\pm$ SD (standard deviation). Levene's and Shapiro-Wilk's tests were used to determine data homoscedasticity and normality, respectively. The significance for hematological and serum biochemical parameters was determined by two factors multivariate analysis (MANOVA) and Tukey's HSD posthoc comparisons to compare the effects of MP among different treatments, exposure times, and their interactions. Statistical analyses were executed using the IBM SPSS Statistics software (version 26). For histopathological alterations, semi-quantitative observations were collected according to Mishra and Mohanty (2008) and Suchana et al. (2021). The histopathological changes in the tissues were determined from an average of the randomly chosen 15 slides from the 45 organ-specific slides. The mean frequency of each histopathological parameter was divided into four categories: no aberrations (–, 0% of segments), mild aberrations (+, < 10% of segments), moderate aberrations (++, 10–50% of segments), and severe aberrations (+++, > 50% of segments).

3. Results

3.1. Uptake of microplastic and the behavior of fish

In the present study, PET fragments were found in the gills, mouth cavity, and digestive tract of *C. batrachus* in the case of T1 (voluntary feeding with PET fragments) and T2 (forced feeding with PET fragments), whereas LDPE pellets were found in the digestive tract in T3 (forced feeding with virgin LDPE pellets) (Fig. S5; Table S1). Overall, the MP-exposed fish ate less, started to swim erratically, and became sluggish. T3 showed lower activity than the other groups (Table S1).

3.2. Changes in hematological parameters

The hematological parameters such as RBCs ($\times 10^6/\text{mm}^3$), WBCs ($\times 10^3/\text{mm}^3$), and hemoglobin (g/dL) of *C. batrachus* on days 30 and 60 of dietary MP exposure are presented in Fig. 1. The values of hemoglobin significantly ($p < 0.05$) decreased in T2 and T3 when compared to control (T0) on days 30 and 60 of exposure. Besides, the values of RBCs ($\times 10^6/\text{mm}^3$) were significantly ($p < 0.05$) decreased in MP-exposed fish when compared to the control on all sampling days. In contrast, the values of WBCs ($\times 10^3/\text{mm}^3$) showed a significant increment ($p < 0.05$) in T1, T2, and T3 compared to control on day 60 of dietary MP exposure (Fig. 1). Table 1 shows two-way MANOVA results for hematological parameters of walking catfish on days 30, and 60 of dietary MP exposure.

3.3. Changes in serum biochemical parameters

The serum biochemical parameters of *C. batrachus* on days 30 and 60 of dietary MP exposure are presented in Fig. 2. Significant ($p < 0.05$) increases in serum glucose (mg/dL) and blood urea (mg/dL) and a notable decrease ($p < 0.05$) in total protein (g/dL) and triglycerides (mg/dL) were found in different treatments on days 30 and 60 of exposure. Moreover, SGPT (U/L) and SGOT (U/L) contents were higher in MP-exposed fish on day 60 of exposure ($p < 0.05$). Table 2 shows Two-way MANOVA results for serum biochemical parameters in fish on days 30 and 60 of dietary MP exposure.

3.4. Structural aberrations of different internal organs

3.4.1. Structural anomalies in the gills

The gills of the control fish had a typical structure with primary and secondary lamellae (Fig. 3), whereas a breakage of lamellae, rupture of the epithelial cell line, epithelial lifting, hyperplasia, hypertrophy of chloride cell, desquamation, necrosis of epithelium, and lamellar fusion were found in the gills of the MP-exposed *C. batrachus* (Fig. 3). Diverse histopathological aberrations found in the gills are summarized in Table 3. On days 30 and 60 of dietary MP exposure, the control group had no apparent histopathological alterations; however, histopathological changes with mild, moderate, and severe conditions were detected in the MP treatment groups (Table 3). The highest frequency of the histopathological changes in gills was found in T1 and, afterward in T2 and T3 groups.

3.4.2. Structural anomalies in the liver

Liver tissue of *C. batrachus* of control (T0) had a typical structure with polygonal hepatocytes (h) (Fig. 4). Macrophage infiltration, fatty degeneration, dilation of blood sinusoids, a clump of pyknotic nuclei, shrinkage of hepatocytes, tilted nuclei, vacuolation, and necrosis was found in the liver of *C. batrachus* exposed to MP (Fig. 4). The histopathological abnormalities found in the liver are shown in Table 4. On days 30 and 60 of dietary MP exposure, histopathological changes with mild, moderate, and severe conditions were noticed in the treatments; however, the control group exhibited no histopathological alterations (Table 4). The highest histopathological alterations in the liver were found in T2, subsequently T3, and T1.

3.4.3. Structural anomalies in the kidney

The kidneys of the control fish had a typical structure with Bowman's capsule, glomerulus, and tubules (Fig. 5). In contrast, occlusion of the tubular lumen, widened intertubular space, deformed tubules, hemorrhage, melanomacrophage center, atrophy of glomerulus, shrinkage of glomerulus were found in the kidneys of *C. batrachus* exposed to MP (Fig. 5). The summary of various histopathological changes in the kidney is provided in Table 5. The highest frequency of histopathological changes in the kidney was found in T2, followed by T3 and T1.

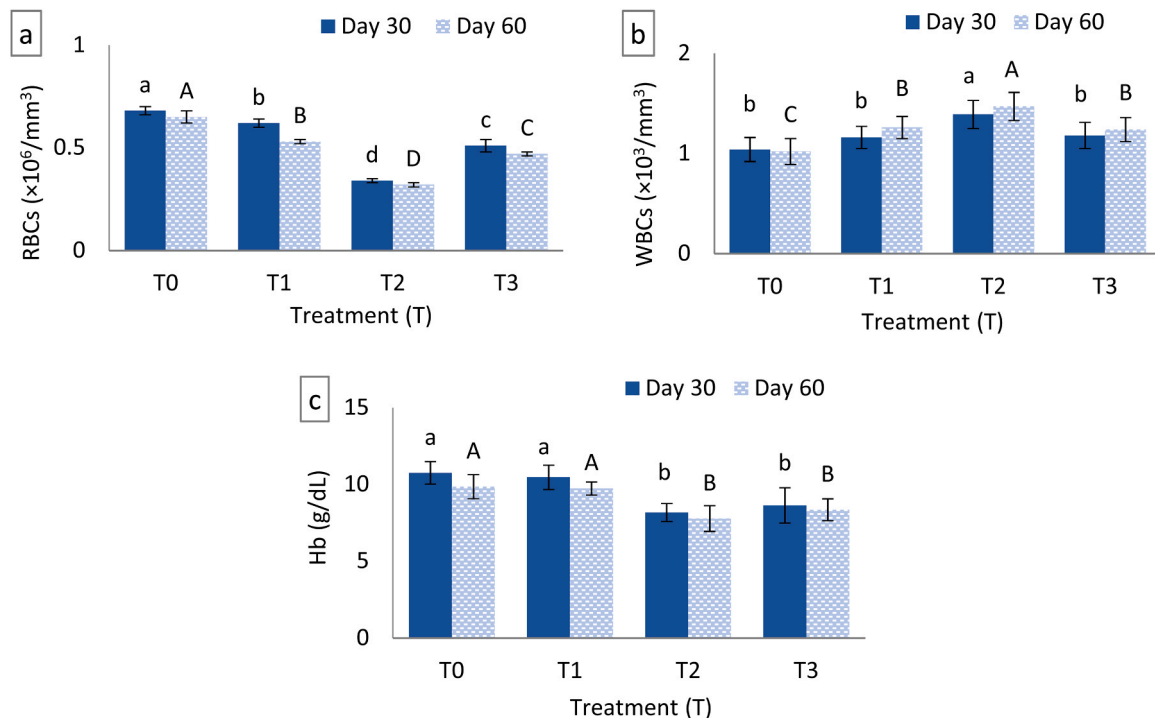


Fig. 1. Changes in hematological parameters of walking catfish at days 30 and 60 of microplastic exposure. Values accompanied by different letters at each sampling day indicate statistically significant differences ($p < 0.05$); Data were analyzed by SPSS program (version 26, IBM, USA).

Table 1

Two-way MANOVA results for hematological parameters of walking catfish on days 30, and 60 of dietary microplastic exposure; p-values computed using $\alpha = 0.05$; p-values < 0.05 and in bold indicate significant differences.

Source	Dependent Variable	df	F	p-value
Treatment	RBCs ($\times 10^6/\text{mm}^3$)	3	421.95	< 0.001
	WBCs ($\times 10^3/\text{mm}^3$)	3	19.67	< 0.001
	Hb (g/dL)	3	26.36	< 0.001
Day	RBCs ($\times 10^6/\text{mm}^3$)	1	49.56	< 0.001
	WBCs ($\times 10^3/\text{mm}^3$)	1	2.56	0.117
	Hb (g/dL)	1	6.50	< 0.05
Treatment*Day	RBCs ($\times 10^6/\text{mm}^3$)	3	5.33	< 0.01
	WBCs ($\times 10^3/\text{mm}^3$)	3	0.52	0.668
	Hb (g/dL)	3	0.41	0.744

df, degree of freedom; RBCs, Red Blood Cells; WBCs, White Blood Cells; Hb, Hemoglobin.

3.4.4. Structural anomalies in the intestine

The intestines of control fish had normal villus, submucosa, muscularis interna, muscularis externa (Fig. 6), whereas the breakage of epithelium, infiltration, rupture of villus, and disruption of submucosa and muscularis layer was found in the intestine of *C. batrachus* exposed to MP (Fig. 6). The histopathological aberrations found in the intestine are cited in Table 6. On days 30 and 60 of dietary MP exposure, histopathological changes with diverse conditions were detected in the MP treatment groups. The highest histopathological changes in the intestine were found in T2, followed by T3 and T1 in terms of severity.

3.5. Changes in growth parameters

C. batrachus exposed to MP showed lower growth performance than the control group ($p < 0.05$) on day 60 (Table 7). Among different treatments, the lowest fish weight gain (%), specific growth rate (% day⁻¹), and survival rate (%) were found in T3, where LDPE pellet-mended feed was fed.

4. Discussion

In the present study, PET fragments were detected in the mouth cavity, gills, and digestive tract of *C. batrachus* in T1 (voluntary feeding with PET fragments) and T2 (forced feeding with PET fragments), while LDPE pellets were observed in the digestive tract of the T3 group (forced feeding with virgin LDPE pellets). The MP-exposed fish gradually ate less and started to swim erratically, and became sluggish, which is almost similar to the effect of other toxic materials in fish, e.g. chromium (Mishra and Mohanty, 2008).

Hematological parameters reveal the physiological and health status of fish, and their investigation is used to assess the influence of environmental toxins (Ahmed et al., 2020; Burgos-Aceves et al., 2018). Hematological properties such as red blood cells (RBCs), hemoglobin (Hb), white blood cells (WBCs), glucose, total protein, triglycerides, urea, serum glutamic pyruvate transaminase (SGPT), serum glutamyl oxaloacetic transaminase (SGOT), etc. are essential biomarkers for estimating the fish health after exposure to several environmental changes, toxicities, microbial contaminations and so on (Kim et al., 2021). Toxic material inflow in the aquatic environment might cause changes in fish hematological characteristics and have a deleterious impact on the oxygen-carrying capability in blood as well as electrolyte balance, resulting in a drop in cell size owing to exosmosis of RBCs (Kavitha et al., 2012; Kim et al., 2021; Narra, 2017; Witeska, 2013). In the present study, it was found that hemoglobin levels and RBCs count were lower in MP exposed fish than in the control group on days 30 and 60 which coincides with Hamed et al. (2022). Choi et al. (2021) suggested that exposure to MP particles can destroy the RBCs membrane, leading to hemolysis and a consequent decrease in hemoglobin concentration. However, the lower levels of hemoglobin and RBC counts in the different treatments of the present study might be due to the damage of the RBCs. Moreover, the values of WBCs were increased in MP-exposed fish groups than in the control on day 60. According to Espinosa et al. (2017), ingesting MP may influence the immune system, affecting animal health and defense. MP may contain, absorb, or exude toxic substances that might impair the immune system (Rochman et al.,

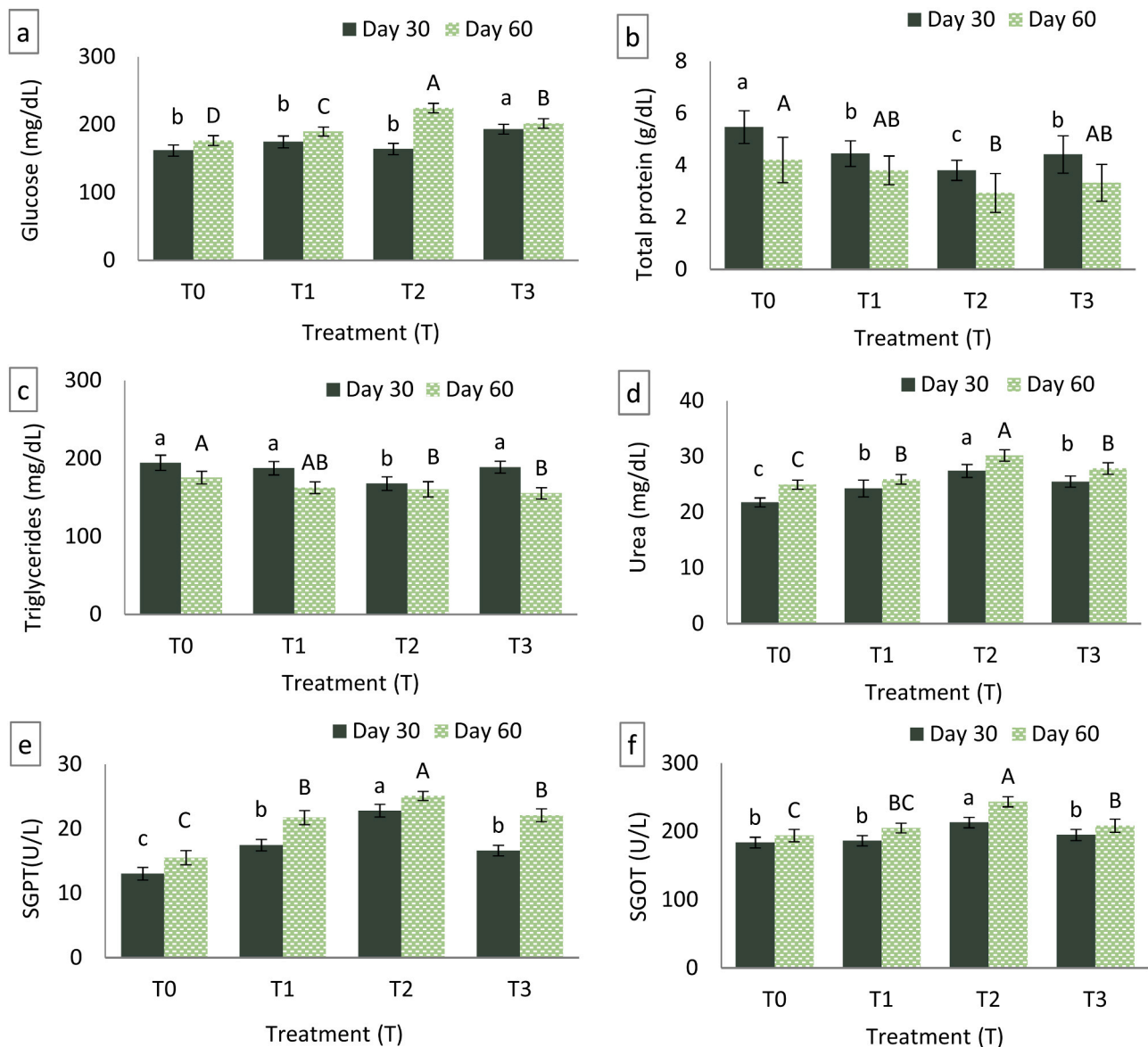


Fig. 2. Changes in serum biochemical parameters of walking catfish at days 30 and 60 of microplastic exposure. Values accompanied by different letters at each sampling day indicate statistically significant differences ($p < 0.05$); Data were analyzed by SPSS program (version 26, IBM, USA).

2013). Blood glucose levels were significantly higher in fish groups exposed to MP than in the control on day 60. MP exposure increases the glucose level in *Oreochromis niloticus* (Raza et al., 2023) and *Clarias gariepinus* (Sayed et al., 2021). Hamed et al. (2019) reported that MP might create hyperglycemia in the blood of Nile tilapia (*O. niloticus*). Blood glucose levels were similarly found to increase in fish after exposure to MP and paraquat (Nematdoost Haghi and Banaee, 2017) and chlorpyrifos and polyethylene glycol (Hatami et al., 2019). Furthermore, hepatic glycogen breakdown produces energy to combat plasticizers' harmful effects and results in elevated glucose levels (Shadegan and Banaee, 2018). Under several physiological circumstances, corticosteroids can regulate fluctuations in glucose levels (Jesop and Hamann, 2004). Therefore, greater corticosteroid levels in fish exposed to MP might have contributed to increased blood glucose. Fish exposed to MP exhibited significantly lower total protein levels than the control groups, matching the study of Nematdoost Haghi and Banaee (2017). Decreased total protein contents in the MP-exposed fish might have reduced protein biosynthesis, which caused liver damage. Proteins and lipids are the most essential organic composites in teleosts, and there is a strong correlation between the proteins (e.g., albumin,

globulin) and lipids (e.g., cholesterol) (Tocher, 2003) following exposure to contaminants (Karami et al., 2016). According to Dere an Dağ (2003) and Ahmad et al. (2010), starvation, impaired protein synthesis in the liver, and problems in protein digestion and absorption may all contribute to a considerable reduction in total protein in fish exposed to MP particles. The interaction of serum proteins with phthalate plasticizers may alter the structure and functionality of all proteins (Yue et al., 2014). In the current study, serum triglyceride levels significantly decreased in MP-exposed fish. A decrease in triglycerides in the blood of fish exposed to paraquat was also reported by Nematdoost Haghi and Banaee (2017). This reduced triglycerides in the blood of fish exposed to MP may be due to disturbance in the absorption of triglycerides in the intestine or disorder in their synthesis in the fish's liver.

Feeding the fish on MP-mended diets led to a significant increment in serum urea levels. The disruption of proteins in an animal's body produces urea, a defense mechanism against nitrogen poisoning. According to Perrault et al. (2017), a good indicator of nephron function might be a change in blood urea concentration. The findings of this study showed that fish's glomerular filtration rate could be affected by MP exposure, which can also damage their renal nephrons. Hamed et al. (2019) found

Table 2

Two-way MANOVA results for serum biochemical parameters of walking catfish on days 30, and 60 of dietary microplastic exposure; p-values computed using $\alpha = 0.05$; p-values < 0.05 and in bold indicate significant differences.

Source	Dependent Variable	df	F	p-value
Treatment	Glucose (mg/dL)	3	34.39	< 0.001
	Total protein (g/L)	3	9.82	< 0.001
	Triglycerides (mg/dL)	3	12.30	< 0.001
	Urea (mg/dL)	3	58.35	< 0.001
	SGPT (U/L)	3	206.10	< 0.001
	SGOT (U/L)	3	53.84	< 0.001
Day	Glucose (mg/dL)	1	125.65	< 0.001
	Total protein (g/L)	1	27.47	< 0.001
	Triglycerides (mg/dL)	1	74.82	< 0.001
	Urea (mg/dL)	1	67.31	< 0.001
	SGPT (U/L)	1	173.23	< 0.001
	SGOT (U/L)	1	59.54	< 0.001
Treatment*Day	Glucose (mg/dL)	3	29.69	< 0.001
	Total protein (g/L)	3	0.47	0.702
	Triglycerides (mg/dL)	3	4.98	< 0.01
	Urea (mg/dL)	3	1.16	0.336
	SGPT (U/L)	3	7.61	< 0.001
	SGOT (U/L)	3	3.54	< 0.05

df, degree of freedom; SGPT, Serum Glutamic pyruvate transaminase; SGOT, Serum Glutamyl oxaloacetic transaminase.

that impaired renal function may have contributed to the rise in urea levels in the blood of MP-exposed tilapia (*O. niloticus*). According to Tomizawa et al. (2015), gastrointestinal bleeding and histopathological impairment can cause a rise in urea levels. Additionally, the primary metabolic processes involved in the urea cycle may be affected by MP, which would increase the amount of urea in the blood (Jin et al., 2019). The cytoplasmic enzymes SGPT (ALT) and SGOT (AST), primarily found in the liver, are crucial for producing glutamate from ketoglutarate (Ellinger et al., 2011). In the present investigation, SGPT and SGOT activity significantly rose in MP-exposed fish on day 60, which agrees with other reported studies (Hamed et al., 2019; Hatami et al., 2019;

Nematdoost Haghi and Banaee, 2017; Shadegan and Banaee, 2018). Increased SGPT and SGOT activity in fish serum may be a sign of tissue injury or inflammation that damages cell membranes and causes the release of these enzymes into the bloodstream (Sinaei and Zare, 2019). The SGPT and SGOT activity can considerably impact intracellular glutamate levels due to the cell membrane's permeability to ketoglutarate (Banaee et al., 2020).

Fish gills are the respiratory organs responsible for several critical functions like osmoregulation, nitrogenous waste excretion, balancing pH, and hormone production (Bernet et al., 1999; Herrero et al., 2018). In the present investigation, the most common change in the gills was the uplifting of the epithelial cell line with enlarged sub-epithelial space. Other than that, changes like lamellar fusion, hyperplasia of epithelial cells, breakage and epithelial sloughing, rupture of the epithelial cell

Table 3

Histopathological alterations in gills of walking catfish exposed to dietary microplastic for different periods.

Aberrations	Exposure days	Treatments			
		T0	T1	T2	T3
Epithelial lifting	30	-	++	++	+
	60	-	++	+++	++
Epithelial hyperplasia	30	-	+++	++	+
	60	-	+++	++	+
Rupture of epithelial cell line	30	-	++	+	-
	60	-	++	-	+
Breakage of lamellae with epithelial sloughing	30	-	+++	+	+
	60	-	++	+	-
Hypertrophy of chloride cell	30	-	++	+	-
	60	-	++	++	+
Lamellar fusion	30	-	+++	+	-
	60	-	+++	+	+
Epithelial necrosis and desquamation	30	-	+	+	-
	60	-	++	++	+

(-) None, (+) mild, (++) moderate, (+++) severe.

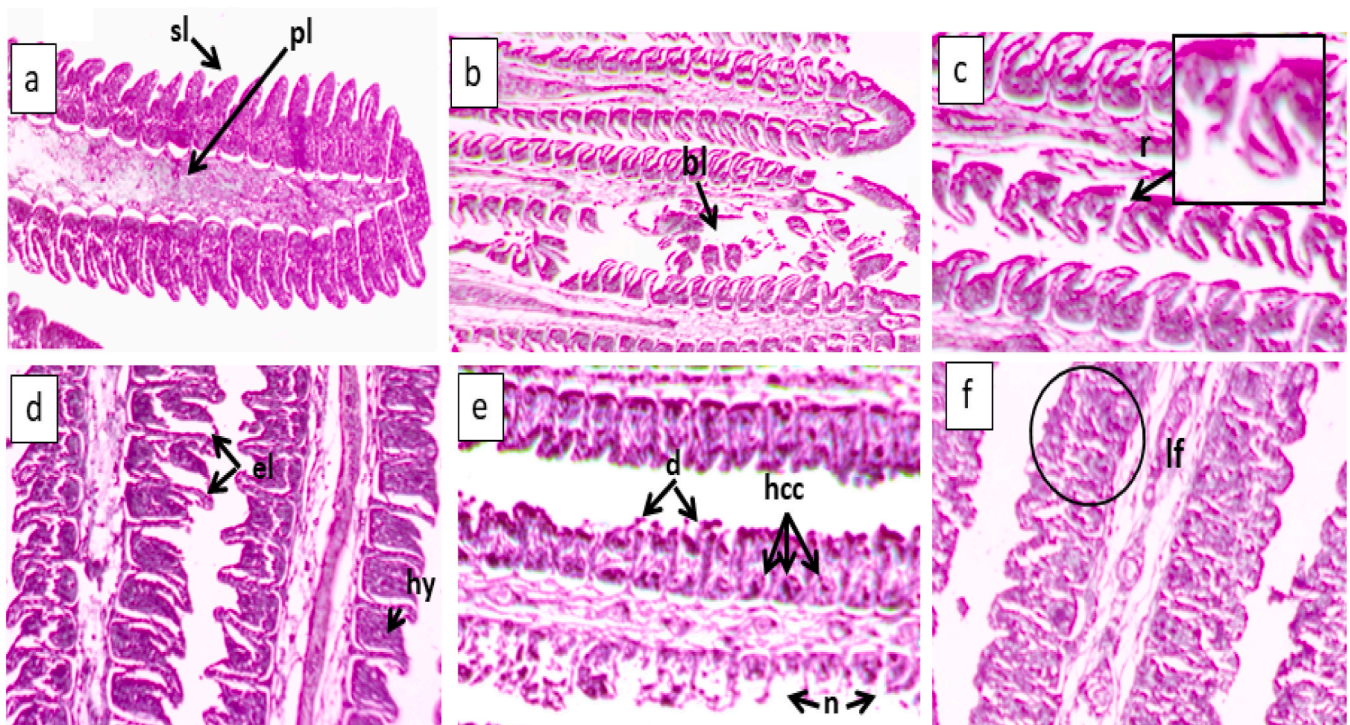


Fig. 3. Histopathological alterations in the gills of walking catfish exposed to dietary microplastic. a. control (10 ×), showing primary (pl) and secondary lamellae (sl); b. breakage of lamellae (bl) (10 ×); c. rupture of epithelial cell line (r) (40 ×); d. epithelial lifting (el), hyperplasia (h) (40 ×); e. hypertrophy of chloride cell (hcc), desquamation (d), necrosis of epithelium (n) (40 ×); and f. lamellar fusion (lf) (40 ×).

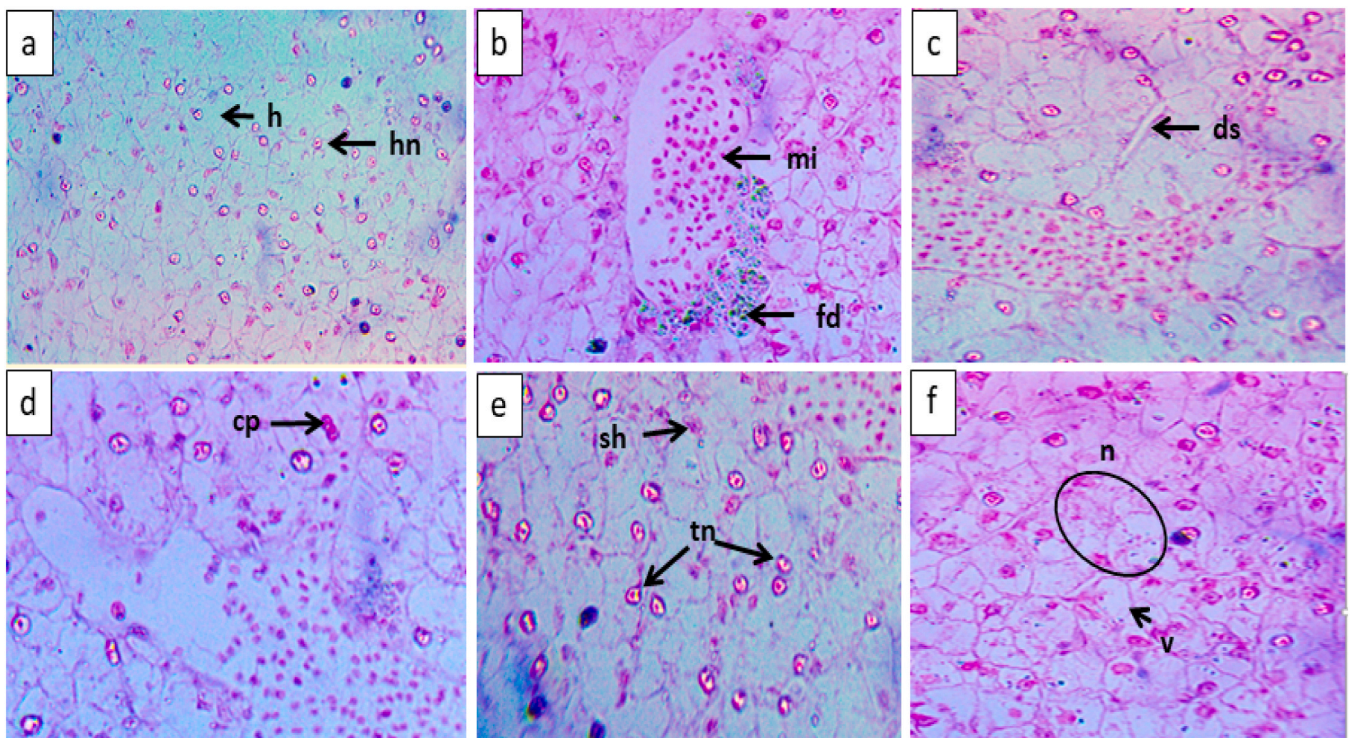


Fig. 4. Histopathological alterations in the liver of walking catfish exposed to dietary microplastic. a. control, showing hepatocyte (h), hepatocyte nucleus (hn) (40 ×); b. macrophage infiltration (mi), fatty degeneration (fd) (40 ×); c. dilation of blood sinusoids (ds) (40 ×); d. clump of pyknotic nuclei (cp) (40 ×); e. shrinkage of hepatocyte (sh), tilted nuclei (tn) (40 ×); f. necrosis (n) and vacuolation (v) (40 ×).

Table 4

Histopathological alterations in liver of walking catfish exposed to dietary microplastic for different periods.

Aberrations	Exposure days	Treatments			
		T0	T1	T2	T3
Macrophage infiltration	30	-	++	++	+
	60	-	+++	+++	+++
Dilated sinusoid	30	-	+	+	+
	60	-	++	++	++
Pyknotic nucleus	30	-	+	++	+
	60	-	++	++	++
Degeneration of hepatocyte	30	-	+	+	+
	60	-	++	+++	+++
Tilted nucleus	30	-	+	++	+
	60	-	++	+++	++
Necrosis and vacuolation	30	-	++	++	++
	60	-	++	++	++

(-) None, (+) mild, (++) moderate, (+++) severe.

line, and hypertrophy of chloride cells were seen in the histopathology of MP-exposed fish. Such deformities were also observed by Karami et al. (2016) and Jabeen et al. (2018) in MP-exposed African catfish *Clarias gariepinus* and goldfish *Carassius auratus*, respectively. These observed effects might have happened due to internal and external mechanical injuries caused by the sharp edges of MP, body immune reactions to leached additives, or joint effects of the two (Hamed et al., 2021; Hu et al., 2020; Karami et al., 2016). The histopathological alterations of gills in the present study were noteworthy in T1 probably because MP-fragments were given separately with commercial feed in T1 (i.e., voluntary feeding), which remained free-floating on the water surface and came in direct contact with the gills of *C. batrachus*, causing the changes. Epithelial sloughing and degeneration are likely due to direct contact with the different sizes and shaped MP (Jabeen et al., 2018). Again, factors like a different size of MP, duration of exposure, their chemical properties, and the target tissue may have induced oxidative

stress and toxicity that resulted in several histopathological changes like abscission, hypertrophy, etc. (Huang et al., 2020; Karami et al., 2016; Wang et al., 2019; Zitouni et al., 2021).

Fundamental metabolic processes carried out by fish livers include the storage and processing of nutrients, the synthesis of enzymes, proteins, and other co-factors, the formation of bile and detoxification of pollutants that enter the fish body, the production of biochemicals required for digestion, and the metabolic activity of xenobiotic compounds (Bernet et al., 1999; Suchana et al., 2021; Wolf and Wolfe, 2005). In our present study, histopathological examination of the livers of fish exposed to diverse types of diet treatments containing MP showed changes like necrosis, vacuolation, pyknotic nucleus, hemorrhage, macrophage infiltration, tilted and merged nuclei, deformed hepatocytes, dilated sinusoids, etc. in the different treatments that coincide with other reported studies (Erkmen et al., 2015; Hamed et al., 2021; Jabeen et al., 2018; Karami et al., 2016). However, dilated sinusoids and macrophage infiltration might have happened due to chemicals released from MP gathered into the liver, which physically congested the hepatic capillaries (Brancatelli et al., 2018). The occurrence of flowing and amassed MP in the liver or the discharge of ethylene monomers from the MP pieces, when they are subjected to proteolytic enzymes and acids, along with microbiological biodegradation, may cause oxidative stress to the liver tissue by producing reactive oxygen species in adherent immune cells (Yang et al., 2014). Hepatocytes are trying to combat this reactive oxygen species, which may be responsible for damaging protein and cell death, resulting in inflammatory responses (Hamed et al., 2021). The histopathological changes found in the livers of *C. batrachus* of the present study have been related to the disturbance of energy metabolism and energy depletion, reduction of protein synthesis in reaction to electrochemical disorders creating chemical stress, the stress of ingesting MP, disaggregation of microtubules, and hepatic erythropoiesis (da Costa Araújo et al., 2020; Hamed et al., 2021; Jabeen et al., 2018).

The kidneys are one of the body's organ systems engaged in

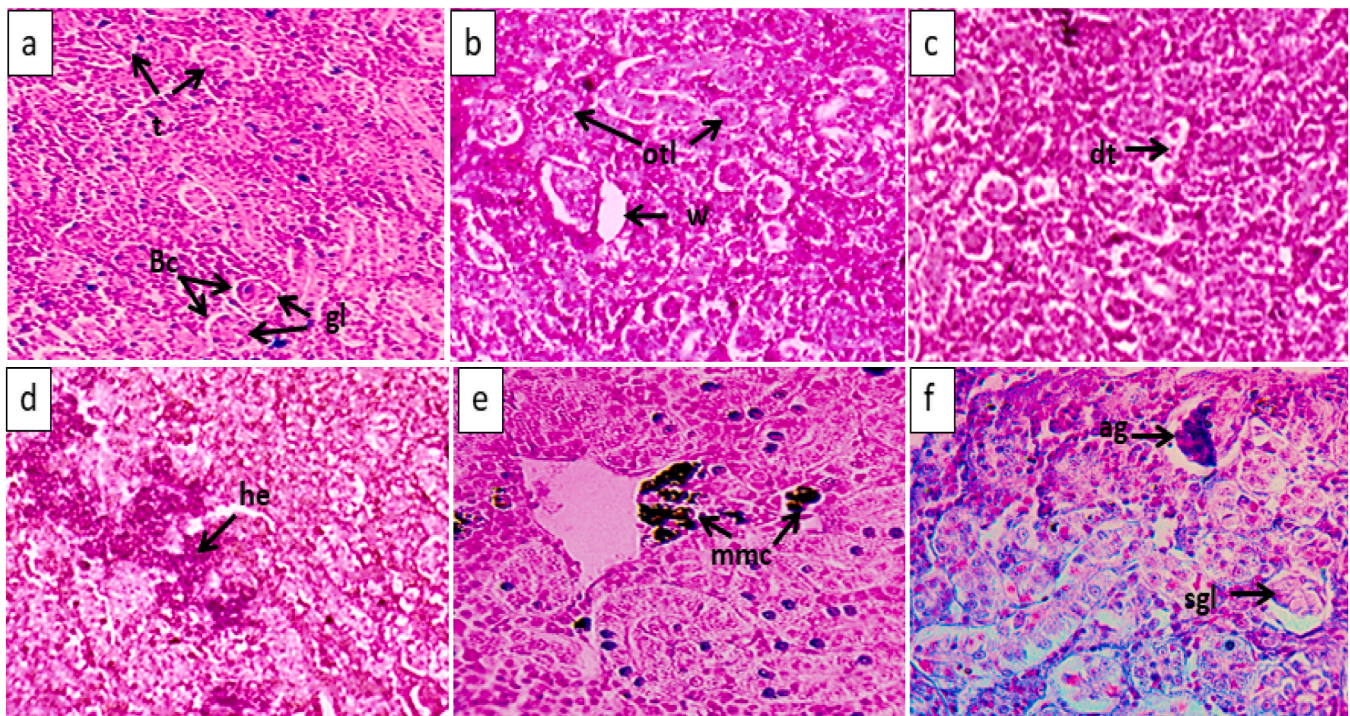


Fig. 5. Histopathological alterations in the kidney of walking catfish exposed to dietary microplastic. a. control, showing Bowman's capsule (Bc), glomerulus (gl), tubules (t) (10 ×); b. occlusion of tubular lumen (otl), widened intertubular space (w) (10 ×); c. deformed tubules (10 ×); d. hemorrhage (he) (10 ×); e. melanomacrophage center (mmc) (40 ×); f. atrophy of glomerulus (ag), shrinkage of glomerulus (sgl) (40 ×).

Table 5

Histopathological alterations in kidney of walking catfish exposed to dietary microplastic for different periods.

Aberrations	Exposure days	Treatments			
		T0	T1	T2	T3
Occlusion of tubular lumen	30	-	+	++	+
	60	-	++	++	+
Atrophy of glomerulus	30	-	-	+	+
	60	-	+	++	++
Shrunken convoluted tubules and degeneration	30	-	+	++	+
	60	-	++	+++	++
Hemorrhage	30	-	+	++	++
	60	-	++	+++	++
Melanomacrophage centre	30	-	++	++	+
	60	-	++	+++	++
Glomerular vacuolation and degeneration	30	-	+	+	+
	60	-	+	++	++

(-) None, (+) mild, (++) moderate, (+++) severe.

excretion and controlling the water balance in fish (Bernet et al., 1999). In the current study, the alterations we found in kidneys were melanomacrophage center, Occlusion of tubular lumen, widened intertubular space due to shrinkage of convoluted tubules, vacuoles, hemorrhage, complete deformation of tubules, inflammatory cell, mitotic nuclei, congested capillaries, atrophy of glomerulus and shrinkage creating larger space between glomerulus and Bowman's capsule, and pyknotic nucleus of fish exposed to various types of diet treatments containing MP. These deformities of the kidney are more or less similar to other published reports (Espinosa et al., 2017; Hamed et al., 2021; Zhu et al., 2020). Additionally, these alterations might have happened due to immune responses of kidneys due to MP associated chemical toxicity effects and sharp edges of fragments, increased blood pressure exerting mechanical pressure on glomerular walls, which might cause injury (Ichimura et al., 2013; Hamed et al., 2021).

The intestine of fish is essential for immunity, endocrine regulating

digestion and metabolism, water and electrolyte balance, and feed digestion and absorption (Le et al., 2019; Schirizzi et al., 2020). In the present study, breakage of epithelium, infiltration, rupture of villus, and disruption of muscularis mucosa layer was found as alterations in MP-exposed fish that support others' findings (Hamed et al., 2021; Jabeen et al., 2018; Pedà et al., 2016). These results might have happened due to different reasons associated with MP. Infiltration and increased space between villi could be related to mechanical abrasion and chemical injury produced by the MP (Espinosa et al., 2019; Hamed et al., 2021). Released ethylene monomers from the MP due to the influence of the digestive enzymes and acids, as well as biodegradation by microbes (e.g., *Enterobacter asburiae* bacteria) present in the gut, might have formed ethylene oxide and ethylene glycol that can render toxic effects in humans and other organisms like fish (Cox and Phillips, 2004; Grosse et al., 2007; Mandal and Ghosh, 2012; Yang et al., 2014). Additionally, internal and external abrasions caused by the sharp texture of the MP might have caused damage to the intestine (e.g., breakage of epithelium, erosion of villi, etc.), reducing nutrient absorption and degrading energy allocation (Cole et al., 2011). Furthermore, the loss of the epithelium layer can allow cytotoxic compounds from the gut lumen to reach the blood, possibly resulting in functional changes in different internal organs (Mobin et al., 2000).

In the present investigation, the MP-exposed *C. batrachus* showed lower growth performance than the control group. Among diverse treatments, the minimum fish weight gain (%), least specific growth rate (% day⁻¹), and lowest survival rate (%) were found in T3 (forced feeding with LDPE pellet), followed by T2 (forced feeding with PET fragments) and T1 (voluntary feeding with PET fragments). However, Critchell and Hoogenboom (2018) and Jabeen et al. (2018) reported that the growth and physical health of fish were negatively impacted when their food was supplemented by plastic. Additionally, ingesting MP might cause false satiety and intestinal damage, which may affect further nutrient absorption and lead to growth inhibition in fish (Lei et al., 2018). Hence, the lower growth performance of *C. batrachus* might have been due to the chemical toxicity of PET fragments and LDPE pellets.

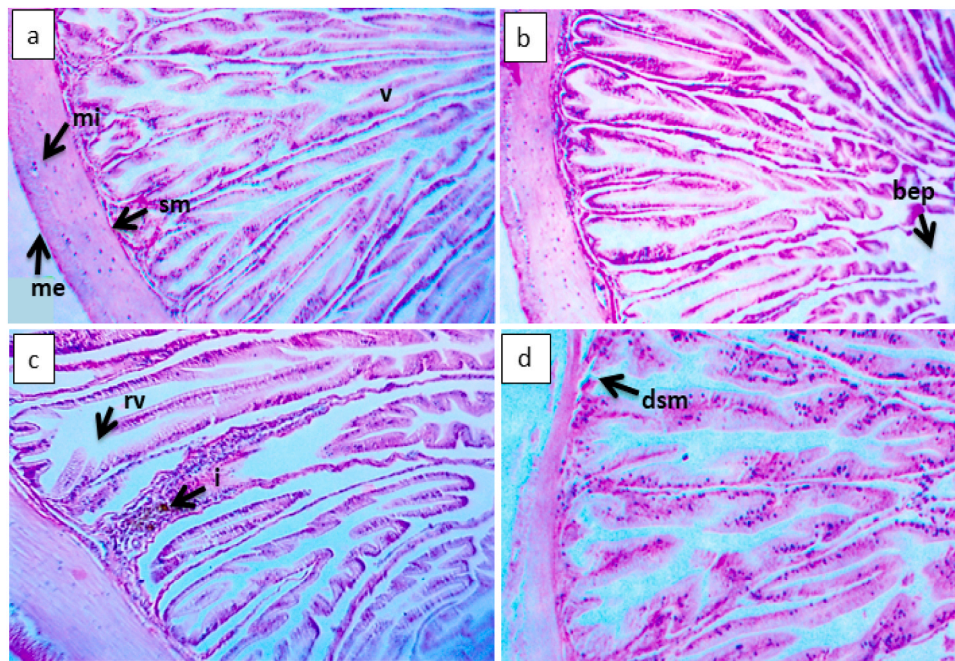


Fig. 6. Histopathological alterations in the intestine of walking catfish exposed to dietary microplastic. a. control, showing villus (v), submucosa (sm), muscularis interna (mi), muscularis externa (me) (10 ×); b. breakage of the epithelium (bep) (10 ×); c. infiltration (i), rupture of villus (v) (40 ×); and d. disruption of submucosa (dsm) (40 ×).

Table 6

Histopathological alterations in intestine of walking catfish exposed to dietary microplastic for different periods.

Aberrations	Exposure days	Treatments			
		T0	T1	T2	T3
Breakage of epithelium	30	-	+	++	+
	60	-	++	+++	++
Infiltration	30	-	+	++	++
	60	-	++	+++	++
Rupture of villi	30	-	+	++	+
	60	-	++	+++	++
Disruption of submucosa and muscularis layer	30	-	+	++	+
	60	-	+	+++	++

(-) None, (+) mild, (++) moderate, (+++) severe.

Table 7

Growth responses of walking catfish in different microplastic-mended feed treatments for two months (60 days).

Growth Parameters	Treatments			
	T ₀	T ₁	T ₂	T ₃
Initial BW (g)	103.33 ± 4.37	102.96 ± 3.85	103.25 ± 4.27	104.08 ± 2.26
Final BW (g)	163.29 ± 3.63 ^a	142.73 ± 5.33 ^b	134.07 ± 3.41 ^b	132.33 ± 4.02 ^b
Fish weight gain (%)	58.15 ± 4.88 ^a	38.63 ± 1.53 ^b	29.93 ± 2.76 ^c	27.13 ± 1.93 ^c
Specific growth rate (% day ⁻¹)	0.76 ± 0.05 ^a	0.54 ± 0.02 ^b	0.44 ± 0.03 ^c	0.40 ± 0.03 ^c
Survival rate (%)	96.67 ± 2.89 ^a	60.00 ± 5.00 ^b	38.33 ± 2.89 ^c	36.67 ± 7.64 ^c

BW, body weight; SGR, specific growth rate, and values with different alphabetical superscripts in a row differ significantly ($p < 0.05$). All values are expressed as mean ± SD.

5. Conclusions

The present study revealed that exposure to MP could result in changes in the various hemato-biochemical parameters and histopathological anomalies, which are sensitive to physiological stressors. Freshwater fish *C. batrachus* were subjected to virgin LDPE pellets and PET fragments associated with their food for 60 days. Hematological indicators, including hemoglobin and RBCs, declined after exposure, but WBCs significantly rose compared to the control. Moreover, we found significant decreases in total protein and triglycerides levels and significant increases in blood glucose, urea, serum glutamic pyruvate transaminase, and serum glutamyl oxaloacetic transaminase activity. The histopathological study of internal organs, including the liver, kidney, intestine, and gills, revealed a number of progressive abnormalities after exposure to MP compared to controls. In addition, *C. batrachus* exposed to MP showed lower growth performance than the control group. Moreover, these further implicate potential negative impacts on fish productivity by causing harm to the vitality of the fish and threatening their lives due to MP pollution with a chemical composition of PET or LDPE. However, the results of this study might support gaining insight into how MP affect aquatic ecosystems. In conclusion, comprehensive studies using molecular biology techniques are required to ascertain the impact of MP on fish and other aquatic biota.

CRediT authorship contribution statement

Kaniz Fatema: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Tasnia Islam Auditi:** Methodology, Formal analysis, Writing – review & editing. **Shema Biswas:** Methodology, Formal analysis, Writing – review & editing. **Sumaiya Binte Ayesha:** Writing – review & editing. **Md. Helal Uddin:** Methodology. **Kizar Ahmed Sumon:** Writing – review & editing. **Chayon Goswami:** Formal analysis, Writing – review & editing. **Ramji Kumar Bhandari:** Resources, Writing – review & editing. **Harunur Rashid:** Conceptualization, Supervision, Writing – review & editing. All authors approved the final version of the manuscript.

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.etap.2023.104250](https://doi.org/10.1016/j.etap.2023.104250).

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